

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Introducing Types of Data Structures



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can read and write about different types of data structures.
- I can write meaningful phrases about types of data structures.
- I can explain different types of data structures.

Ask your partner:

- What do you know about the way data is organized in companies?
- Do you store any type of data in the same way? Why? Why not?





Let's Explore!

Today, I'm attending a lecture about types of data structures.



Instructions: Read the text below.

Professor Moore: Good evening, everyone! Thanks for coming to this lecture about data structures. In this opportunity, we are discussing different types of data structures. You're going to learn valuable information, so I suggest taking notes. Let's start. As you know, a data structure is the particular organization of data in a computer program. Depending on the type of information your computer needs to categorize, you are going to use a different type of data structure. The two common types computer science is using nowadays are arrays and types of lists.

Concerning the arrays, they are basic data types that haven't any internal links between their elements, so it's necessary that all data belongs to the same type. It stores data in addressable locations sequentially. In other words, the computer program is storing and retrieving data items constantly. On the other hand, regarding the types of lists, unlike the arrays, they have one or more pointers to link their elements. As their name suggests, there is a strong tendency to work with specialized lists in this type of data structures. Among the types of specialized lists, we have trees, stacks, and queues.

A tree list consists in carrying out the data organization in a hierarchical order through a root node and sub nodes. When your computer is executing this type of list, it speeds important functions like "inserting" or "deleting". With regard to stacks, they work according to the order of access, or the LIFO order (last in, first out). It means the last item the program inserts is going to be at the top, removing the previous item. On the contrary, queues execute the order FIFO (first in, first out); that's to say, the last item the program inserts is going to be at the end, and the one to solve is going to be the first in the queue. This type of list plays a part when there are specific tasks that need special attention.

To conclude, I invite you to summarize the most important aspects of this lecture in some meaningful phrases. I really enjoyed giving this lecture tonight. Keep reading about arrays and lists. You can also practice surfing the web to get more examples and information.



Let's Practice!

Which is better, an array or a list?

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. When you work with a list, you are going to have the possibility to work with specialized lists.
2. It's necessary that all data belongs to the same type.
3. The computer program is storing and retrieving data items constantly.
4. It stores data in addressable locations sequentially.
5. They have one or more pointers to link their elements.
6. Nowadays, we are working with trees, stacks, and queues.

Array	List



Let's Practice!

There's a lot of information about data structure.

Instructions: Find the words in the word search below.

1. Depending on the type of information your computer needs to categorize, you are going to use a different type of data _____.
2. The two common types computer science is using nowadays are _____ and types of lists.
3. When your computer is executing the tree list, it speeds important functions like "_____" or "deleting".
4. Stacks work according to the order of access, or the _____ order (last in, first out).
5. Queues execute the order _____ (first in, first out); that's to say, the last item the program inserts is going to be at the end.

V	D	E	G	E	L	I	V	I	R	P	R	E	D	N	U
E	Q	N	B	O	L	S	I	T	A	U	U	H	G	H	M
L	O	F	N	O	I	T	A	T	C	E	P	X	E	S	E
J	C	R	S	P	G	C	L	L	D	T	B	G	G	C	U
V	Z	E	T	S	V	T	I	C	B	S	I	N	E	L	I
D	M	S	R	Z	M	W	F	G	A	K	D	I	E	C	O
S	K	I	U	R	Y	G	O	E	K	L	G	T	V	K	T
S	E	T	C	V	A	T	E	S	O	V	L	R	I	X	I
O	M	A	T	D	L	F	L	E	K	B	Q	E	T	K	L
T	U	S	U	C	O	B	D	M	V	Q	O	S	I	B	L
I	X	M	R	I	C	I	T	O	S	I	N	N	N	R	I
N	X	G	E	T	E	O	V	T	R	H	E	I	I	R	C
A	O	S	U	F	L	O	R	F	I	F	O	A	S	P	O
N	E	P	H	A	R	R	A	Y	S	Y	I	M	E	V	P
A	E	L	Y	U	Q	U	I	T	A	V	L	Z	D	X	E
B	M	B	U	Z	D	N	A	H	G	N	I	P	L	E	H



Extension Activity

What data structure is more common?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search about the types of data structure that companies use in your country.
- Make a list of 5 companies with their respective structures.
- Make a poster showing what companies do to organize their data.
- Use the grammar and vocabulary you studied in this learning object.
- Present your work to your class.

Self-Assessment

- I look for the types of data structure that companies normally use in my country.
- I recognize the respective data structures that some companies use.
- I create a poster summarizing all the information I found.
- I use the grammar and vocabulary I learned.
- I talk about the data structures companies use in front of my classmates and teacher.

Yes

No

☐ ☐
☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- **internal link** (noun): hyperlink that points at any target
e.g. Concerning the arrays, they are basic data types that haven't any internal links between their elements.
- **pointer** (noun): something that directs or indicates the location of data items in a data structure
e.g. Regarding the types of lists, unlike the arrays, they have one or more pointers to link their elements.
- **root node** (noun): the principal part in a tree list
e.g. A tree list consists in carrying out the data organization in a hierarchical order through a root node and sub nodes.
- **store** (verb): to enter or retain data
e.g. It stores data in addressable locations sequentially.
- **retrieve** (verb): to find and return data
e.g. The computer program is storing and retrieving data items constantly.
- **queue** (noun): a line of specific tasks waiting for the computer system to solve them
e.g. The last item the program inserts is going to be at the end, and the one to solve is going to be the first in the queue.



Learning Points

Simple Present, Present Continuous, Be Going to in Affirmative

You use the **simple present tense** to talk about routines or facts. You use the **present continuous tense** to describe actions that are developing at a particular period of time. Finally, you use **be going to** to express actions in a near future.

Simple present tense

You use this tense to talk about things that you generally do, like your daily routines. You also use this tense to talk about general truths or facts.

Structure:

Subject + **verb*** + complement

*Remember that for the third person of the singular (he, she, and it), you need to add an "s" "es" or "ies" to the verb.

Examples:

A *stack list* **works** according to the order of access, or the LIFO order.

On the contrary, queue lists **execute** the order FIFO.

Present continuous tense

You use this tense to talk about things that are happening at the moment we are speaking.

Structure:

Subject + **to be verb** + **main verb** + **ing** + complement

Example:

My computer system *is* **organizing** data in a stack list.

Be going to

You use this tense to express future plans and intentions.

Structure:

Subject + **to be verb** + **going to** + **main verb in base verb** + complement





Learning Points

Example:

A queue list **is going to work** through a FIFO order.

Examples

- A tree list consists in carrying out the data organization in a hierarchical order through a root node and sub nodes.
- In other words, the computer program is storing and retrieving data items constantly.
- It means the last item the program inserts is going to be at the top, removing the previous item.

Verb + Ing

In some cases, the verb that goes after another verb may go in the **-ing form**.

The following chart summarizes the specific cases in which you use the verb + ing form:

Verb + Ing Common Uses		
Verbs to show like or dislike	Verbs of thinking and saying	Other usual verbs
Like Dislike Love Hate Enjoy	Consider Suggest Admit Remember Imagine	Avoid Keep Start Practice Finish
Examples: <i>I enjoy learning about different types of data structures.</i>	Examples: <i>The engineer suggests using a tree list to store this type of data.</i>	Examples: <i>Practice looking for other types of data structures.</i>

Examples

- I really enjoyed giving this lecture tonight.
- Keep reading about arrays and lists.
- You can also practice surfing the web to get more examples and information.



Learning Points

Academic Collocations

An academic collocation is the union of some words related to a specific context that are often used and sound natural in the English language.

You can use some particular academic collocations to describe your work with data structures. Take a look at the following chart:

Academic Collocations in Software	
Collocation	Meaning
Strong tendency	to decide on something above other possibilities; to choose
Gather evidence	to collect relevant information about a particular topic
Carry out	to execute something; to accomplish a task
Play a part	to play an important role in something
Supporting evidence	to provide useful information to prove something

Examples

- *There is a strong tendency to work with specialized lists in this type of data structures.*
- *This type of list plays a part when there are specific tasks that need special attention.*